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# Islamic University of Technology

Department of Computer Science and Engineering (CSE)

B.Sc. in Software Engineering

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## Project Proposal

TypeDrift

### Group 2

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# TypeDrift

## 1 Project Overview

Typing proficiency has become an essential skill in the modern digital landscape, impacting productivity across academic, professional, and personal domains. Whether taking lecture notes, writing reports, coding, or communicating via email, efficient typing directly translates to time savings and improved work quality. However, developing this skill requires deliberate practice in an environment that provides structured feedback and tracks progress over time.

Despite the abundance of typing practice tools available, most existing solutions suffer from significant limitations. Web-based platforms dominate the market but require constant internet connectivity, are often cluttered with advertisements, and lack comprehensive analytics for long-term progress tracking. For users who work primarily in terminal environments—such as developers, system administrators, and programming enthusiasts—the absence of a lightweight, CLI-based typing tutor is particularly frustrating.

To address these challenges, we are developing TypeDrift—a Command-Line Interface CLI-based typing performance monitor designed to provide a streamlined and efficient typing practice experience. Built entirely from scratch using C++, TypeDrift operates completely offline and focuses on improving users' typing speed and accuracy.

TypeDrift provides real-time performance metrics during practice sessions, including Words Per Minute (WPM), accuracy percentage, and elapsed time. It supports multiple practice modes and difficulty levels, maintains detailed session histories for progress tracking, and includes a local leaderboard and mini-game mode to enhance motivation. Security is ensured through password hashing, and all user data is stored locally to preserve privacy.

## 2 Motivation Behind the Project

The motivation for developing TypeDrift arises from the widespread issue of typing inefficiency among students and professionals. In academic settings, slow typing hinders note-taking and assignment completion, while in professional environments it reduces productivity during tasks such as writing reports, responding to emails, or coding. Despite this need, many users lack access to effective, convenient, and distraction-free typing practice tools.

Although numerous typing applications exist, many suffer from serious limitations. Most web-based platforms require constant internet access and are cluttered with advertisements or distracting interfaces. Furthermore, they often fail to provide meaningful analytics or long-term progress tracking, preventing users from accurately measuring improvement.

For developers and terminal-focused users, the lack of a lightweight CLI-based typing tutor is especially problematic. Switching to graphical applications disrupts workflow, whereas a CLI tool integrates seamlessly into terminal environments and supports focused practice.

TypeDrift addresses these issues by offering an offline, CLI-based typing practice solution with persistent performance tracking. It also incorporates gamification features such as a local leaderboard and mini-game mode to encourage consistent practice, while storing all user data locally to ensure privacy.

### 3 User Requirement

Through discussions with potential users and analysis of existing typing practice tools, we have identified the following key requirements for TypeDrift:

#### 1. User Authentication and Account Management

Users need the ability to create personal accounts with secure credentials and log in to access personalized features. The system must implement password hashing to protect user data and prevent unauthorized access.

#### 2. Multiple Practice Modes and Difficulty Levels

Users require flexibility in choosing how they practice, including passage-based drills for accuracy enhancement alongside more engaging game modes. The system should offer multiple difficulty levels to accommodate users ranging from beginners to advanced typists.

#### 3. Real-time Performance Feedback

During practice sessions, users need immediate visual feedback on their typing performance. The system must display live metrics including WPM, accuracy percentage, and elapsed time, while providing color-coded feedback to distinguish correct input, errors, and untyped text.

#### 4. Comprehensive Session Reports and Analytics

After each practice session, users need detailed performance summaries showing key metrics such as final WPM, accuracy, error count, and time taken. The system should maintain a complete history of all sessions for long-term progress tracking.

#### 5. Personal Statistics and Progress Tracking

Users need access to their historical performance data to monitor improvement over time. The system should provide visualizations and summaries of WPM trends, accuracy progression, and overall practice statistics.

## 6. Competitive Leaderboard System

To foster motivation and engagement, users require access to a local leaderboard that ranks participants based on their peak WPM performance, enabling them to compare their skills with others.

## 7. Complete Offline Functionality

Users need the ability to access all features, practice typing, view statistics, and manage accounts without requiring an internet connection. All data must be stored locally to ensure privacy and enable offline operation.

## 8. Engaging Mini-game Mode

To maintain user interest and prevent practice fatigue, the system should offer an entertaining mini-game mode that provides variety while still improving typing skills.

# 4 Key Features

The application's key features are organized into two distinct categories: **Main Features**, which form the core interactive experience, and **Side Features**, which provide supplementary utility and engagement. Both are detailed in the sections below :

## Main Features

- **User Authentication System**

Secure sign-up and login functionality with password hashing using cryptographic algorithms. User credentials are stored locally in encrypted format, ensuring data privacy and protection against unauthorized access.

- **Multiple Practice Modes**

Passage-Based Drills where users complete entire passages focusing on accuracy and consistency alongside some interesting challenges. Where user has to type accurately and skillfully, forcing the user to think while typing.

- **Difficulty Level Selection**

Three difficulty tiers for varying skill levels: Beginner (easy), Intermediate (mixed words with moderate complexity), and Advanced (technical terms, complex punctuation, and challenging text structures).

- **Real-time Performance Metrics**

Live display of key performance indicators during practice sessions including: (a) Words Per Minute (WPM) - calculated dynamically based on correct characters typed, (b) Accuracy Percentage - ratio of correct to total characters, (c) Time Elapsed - current session duration, and (d) Error Count - number of mistakes made.

- **Color-coded Visual Feedback**

Intuitive color system to enhance user experience: (a) Yellow text for correctly typed characters, (b) Red text highlighting typing errors, and (c) Grey text showing remaining untyped portions of the passage.

- **Detailed Session Reports**

Comprehensive post-session summaries displaying: (a) Final WPM score, (b) Overall accuracy percentage, (c) Total time taken, (d) Number of errors made, and (e) Performance comparison with previous sessions.

- **Personal Statistics Dashboard**

Historical performance tracking showing: (a) WPM progression over time, (b) Accuracy trends across multiple sessions, (c) Total practice time accumulated, (d) Best performance records, and (e) Average performance metrics.

## Side Features

- **Session History Viewer**

Complete log of all previous practice sessions with filters and search capabilities, allowing users to review specific sessions and identify patterns in their performance.

- **Mini-game Mode**

Engaging typing challenges designed to reduce monotony while improving skills through gamified exercises such as word races and accuracy challenges with score tracking.

- **Local Leaderboard System**

Competitive ranking feature displaying: (a) Top performers ranked by peak WPM, (b) Username and highest score for each user, (c) Current user's position and score, and (d) Automatic updates after each session.



## 5 Flow Chart / Class Diagram

The system architecture and program flow are depicted in the flowchart below. This diagram, presented in Figure 1, illustrates the core logic, decision points, and component interactions within the application.

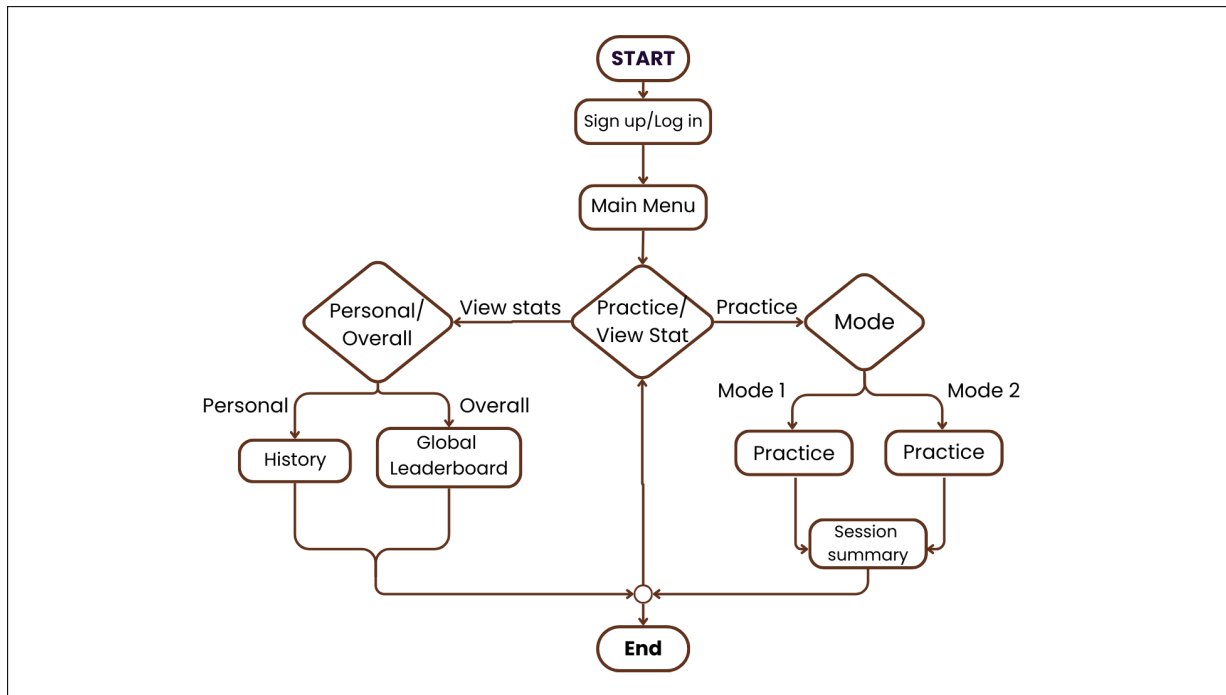


Figure 1: Application Flow Chart

### Application Flow Description

**Start:** The application initializes, loading user data and configuration files into memory.

**Authentication Menu:** Users can Sign Up or Log In; credentials are validated and securely stored using password hashing.

**Main Menu:** After authentication, users choose to start a practice session, view statistics, view the leaderboard, check session history, or exit.

**Start Practice Session:** Users select a practice mode (Timed or Passage-based) and a difficulty level, after which relevant text is loaded.

**Practice Session Execution:** Users type the displayed text while the system tracks input in real time and shows live WPM, accuracy, and time.

**Session Completion:** Upon completion, final metrics are calculated and a session summary is generated.

**Save Session Data:** Results are saved to the user profile, updating statistics and the leaderboard if a new best score is achieved.

**View Statistics:** Users can view summarized performance data including WPM, accuracy trends, and total practice time.

**View Leaderboard:** Displays user rankings based on highest WPM to encourage competition.

**Session History:** Users can review brief records of past sessions with key performance details.

**Return to Main Menu:** Users may return to the main menu after completing any action.

**Exit:** All data is saved locally, and the application exits.

## 6 Tools and Technologies

The project is being developed as a headless CLI application using C++ as the core programming language. We are using Visual Studio Code (VS Code) as our primary Integrated Development Environment (IDE) for its lightweight nature and strong C++ support via extensions. Version control is managed through Git, with the source code hosted on GitHub to facilitate collaboration, code review, and submission as mandated by the course policy.

No external frameworks or game engines are used; the implementation is done from scratch with minimal standard library usage.

### Technology Stack

Table 1: List of Tools and Technologies

| Category             | Tool / Technology            |
|----------------------|------------------------------|
| Programming Language | C++ (Standard Library only)  |
| IDE                  | Visual Studio Code (VS Code) |
| Version Control      | Git                          |
| Hosting Platform     | GitHub                       |

## 7 Proposed Timeline

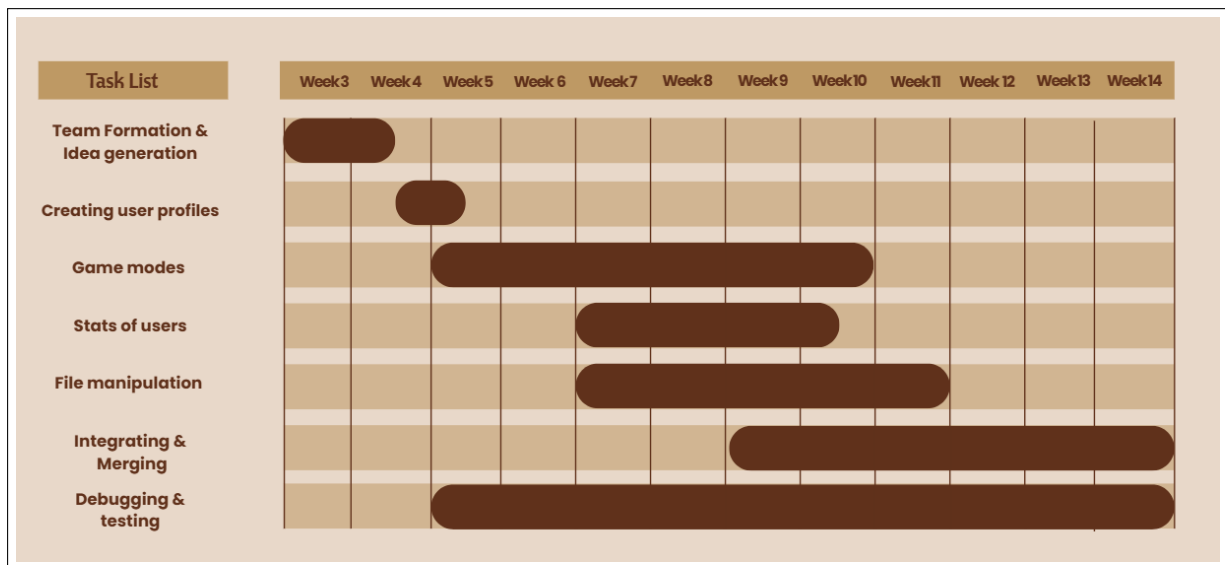


Figure 2: Timeline Diagram (Gantt Chart)

## 8 Suggestions Received

### 1. Adding hashing method for storing passwords

*Response:* This suggestion is already under implementation. We will use a cryptographic hash function to securely store user passwords, ensuring that plain-text credentials are never saved in the database.

### 2. Using Grey-text while playing the game

*Response:* This suggestion has been considered as well. Untyped portions of text will be displayed in grey, while typed characters will show in yellow (correct) or red (errors), providing clear visual feedback.

### 3. Implementing local leaderboard

*Response:* This suggestion has been accepted and included as a main feature. A local leaderboard will rank users based on their peak WPM performance.

### 4. Custom test practice for each users

*Response:* For now we are not considering this feature since it doesn't align with our initial scope of the project.

## 9 Links

1. Presentation Slide Link[click on text]
2. GitHub Repository Link[click on text]